Project Title: Analyzing the Drivers of CO_2 Emissions

Course: DAB422 – Capstone Project 2
Instructor: Dr. Manjari Maheshwari
Group Name: 12
Group members:

- Ravina Ravina – 0874291
- Riya Rani – 0867993
- Dinesh Reddy Dekkatha – 0876580
- Brien Chijioke Ihuka – 0880506
- Mohammad Sohel Ahmed – 0875336

Institution: St. Clair College, Windsor

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https://github.com/RiyaRani12/Capstone-Project

Table of Contents

Table of Contents

1. **Title Page** 1
2. **Table of Contents** 2
3. **Abstract** 3
4. **Introduction** 4
5. **Data Preparation** 5
6. **Methodology and Implementation**..... 6
7. **Optimization** 7
8. **Results and Evaluation** 8
9. **Model Deployment**……9
10. **Discussion and Key Insights**……...10
11. **Conclusion**……..11
12. **Suggestions**……..12
13. **Future Work**……..13
14. **References**……..14

Abstract

This project focuses on analyzing the key factors influencing CO_2 emissions using data analytics and machine learning techniques. Initially, CO_2 emissions data was analyzed, and later renewable energy consumption data was incorporated to better understand their relationship.

Various machine learning models, including Linear Regression, Decision Tree, Random Forest, and Artificial Neural Network (ANN), were applied and compared. Among these, Random Forest achieved the highest accuracy and demonstrated the best performance.

The results indicate that economic factors, particularly CO_2 per GDP, have the strongest impact on emissions, while renewable energy shows a relatively weak direct relationship.

Overall, the study highlights the importance of improving economic efficiency and reducing carbon intensity to effectively control CO_2 emissions.

Introduction

In today's world, climate change has become one of the most critical global challenges. One of the major contributors to climate change is the increase in carbon dioxide (CO_2) emissions. Governments and organizations are continuously working towards reducing emissions, but the key factors driving these emissions are not always clearly understood.
In this project, I analyzed the major drivers of CO_2 emissions using data analytics and machine learning techniques. The goal was to understand whether renewable energy plays a significant role in reducing CO_2 emissions and identify the most influential factors.

Project Overview

The project initially focused on identifying the countries with the highest CO_2 emissions. However, this approach did not provide a fair comparison, as countries with larger populations naturally showed higher emissions. To address this, the analysis was refined to focus on **CO_2 emissions per capita**, which allowed for a more accurate and meaningful comparison across countries.
The main objective of this project is:

- To identify key factors driving CO_2 emissions
- To analyze the relationship between renewable energy and emissions
- To use machine learning models for prediction and analysis

Key Questions

1. What are the main factors influencing CO_2 emissions?
2. Is renewable energy effective in reducing emissions?
3. Which machine learning model provides the best prediction?

Data Source

The dataset used in this project was obtained from the World Bank

Initially, a dataset containing CO_2 emissions data from 1960 to 2022 was used to identify countries with high emission levels. During the analysis, it was observed that some countries had significantly higher CO_2 emissions.

To further understand this pattern, an additional dataset on renewable energy consumption (1960–2022) was included. This was done to analyze the relationship between renewable energy usage and CO_2 emissions, and to determine to what extent renewable energy can help in reducing emissions.

Both datasets were sourced from the World Bank and were later cleaned and merged to form a single comprehensive dataset. This final dataset was then used for further analysis

Data Cleaning and Preparation

Before starting the analysis, both datasets required extensive cleaning and preprocessing.

Initially, each dataset was cleaned separately. There were a large number of null values, which were handled using median imputation to maintain data consistency without affecting the overall distribution.

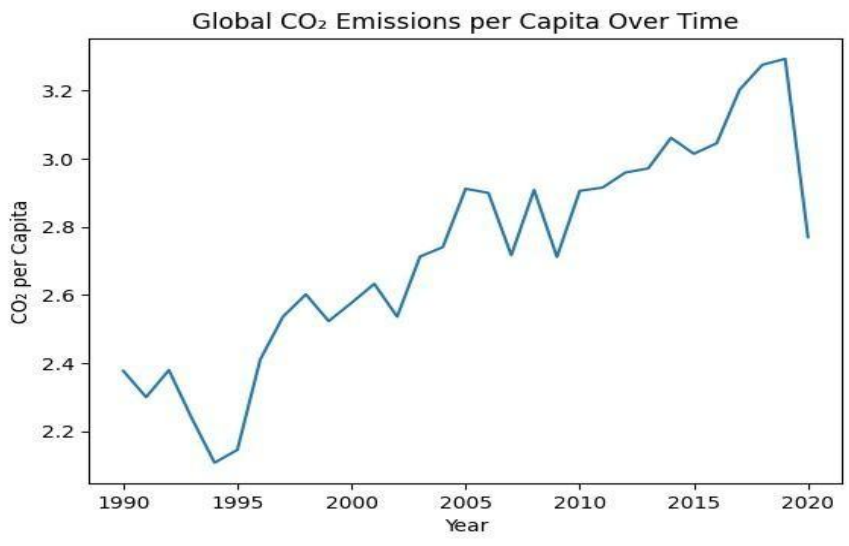
Additionally, the datasets contained many small territories and irrelevant regions, which were removed to ensure meaningful country-level analysis.

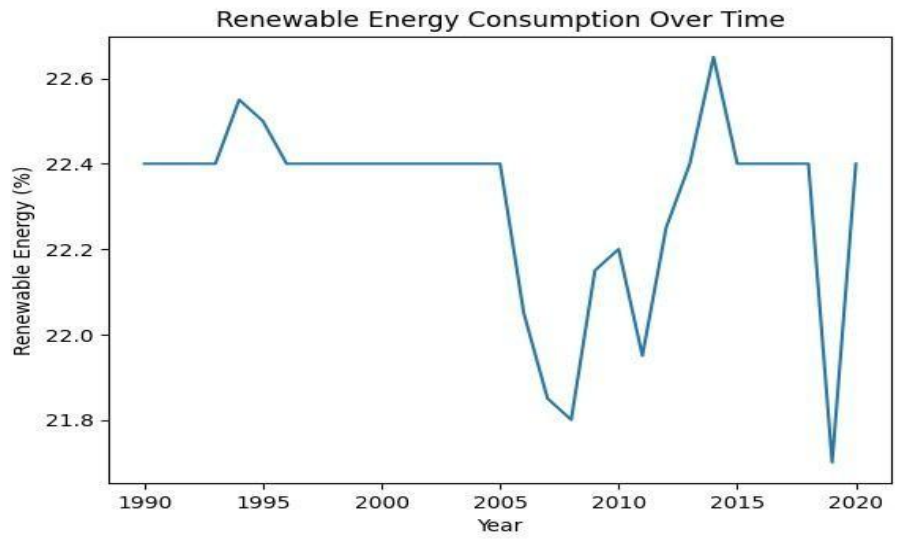
After cleaning both datasets individually, they were merged into a single dataset. Following the merge, the combined dataset was carefully reviewed again to check for any inconsistencies or remaining missing values.

Further preprocessing steps included converting all variables into the appropriate numeric format and filtering the data for the time period 1990 to 2020 to ensure consistency across all features.
After completing these steps, the final dataset contained data for approximately 198 countries, which was then used for further analysis.

Exploratory Data Analysis

CO_2 vs Renewable Energy





Key Findings

- CO_2 emissions show an **overall increasing trend** over time
- Renewable energy consumption remains **relatively stable with minor fluctuations**
- There is **no strong inverse relationship** between renewable energy and CO_2 emissions
- This suggests that **renewable energy alone is not sufficient** to significantly reduce CO_2 emissions

Methodology and Implementation

Models used in this project include Linear Regression, Decision Tree, Random Forest, ANN (Artificial Neural Networks, and Gradient Boosting. The dataset was divided into training and testing sets to evaluate performance.

Baseline Models

Model	R^2 Score	MAE	RMSE
Linear Regression	0.568	9.223	40.793
Decision Tree	0.999	0.514	1.575

Train vs test

Model	Train R^2	Test R^2
Linear Regression	0.666	0.568
Decision Tree	1.000	0.999

Advance Models

Model	R^2 Score	MAE	RMSE
Random Forest	0.9988	0.649	2.114
Gradient Boosting	0.9972	2.010	3.263

Train vs test

Model	Train R^2	Test R^2
Random Forest	0.9997	0.9988
Gradient Boosting	0.9987	0.9972

In this project, multiple machine learning models were implemented and compared in two stages: **baseline models** and **advanced models**.
The **baseline models** used in this study include: **Linear Regression** and **Decision Tree**

Linear Regression showed moderate performance with an R^2 score of around 0.568, indicating that it was not able to capture complex relationships in the data.On the other hand, Decision Tree achieved a very high R^2 score (close to 0.999) on training data. However, such high performance may indicate overfitting, meaning the model may not generalize well to unseen data.
Linear Regression showed consistent performance with similar train and test R^2 scores (**0.666 train, 0.568 test**), indicating good generalization

Decision Tree showed extremely high accuracy (**1.000 train, 0.999 test**), which suggests a high risk of **overfitting**. Therefore, **Linear Regression was selected as the final baseline model** due to its more stable and reliable performance.

The **Advanced models** used were: **Random Forest** and **Gradient Boosting**

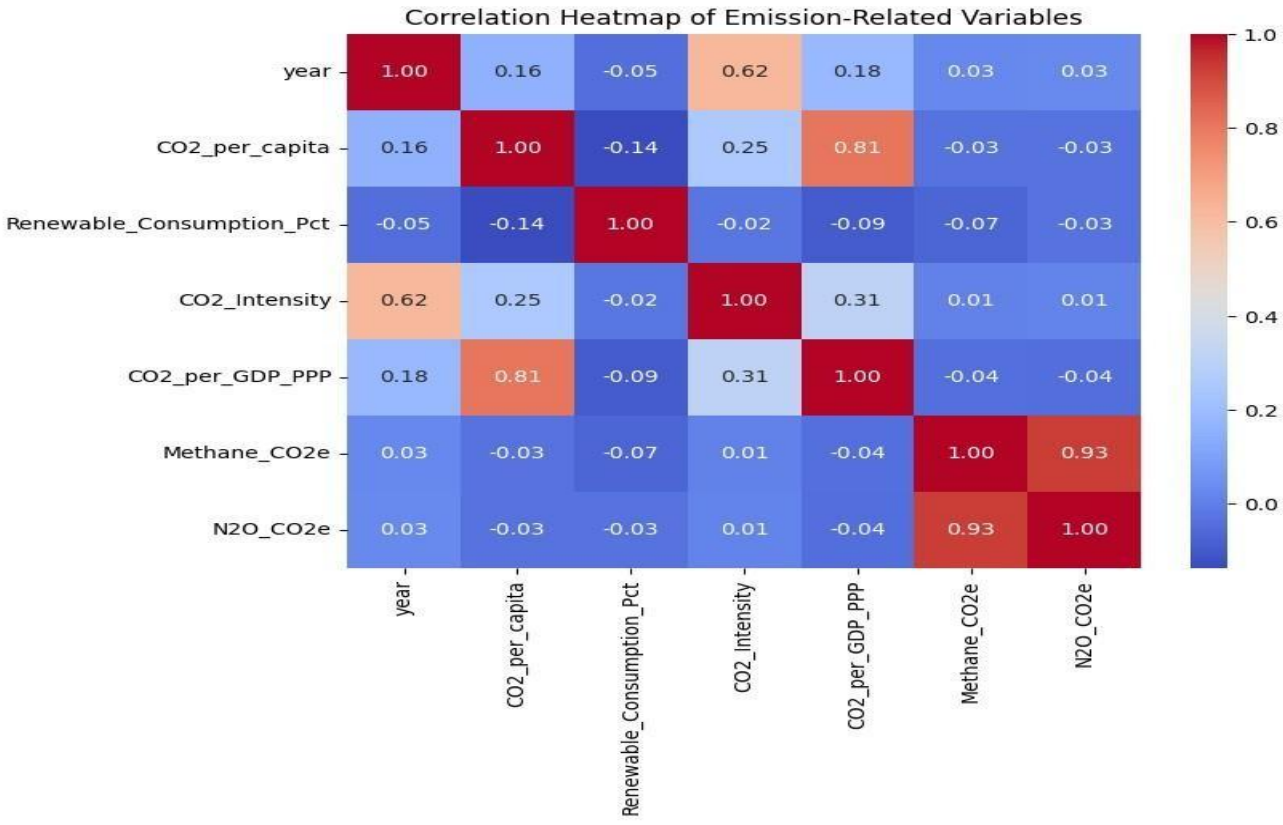
These models were applied to improve prediction accuracy and capture complex relationships in the data.

Random Forest achieved the highest performance with an R^2 score of approximately **0.9988**, **Gradient Boosting** also performed well but slightly lower than Random Forest.

Random Forest achieved very high and balanced performance (**0.9997 train, 0.9988 test**), showing both high accuracy and good generalization. **Gradient Boosting** also performed well but slightly lower than Random Forest

After comparison, **Random Forest was selected as the final advanced model** due to its superior accuracy and lower error values. Further we will compare our final model with ANN model.

Correlation Analysis

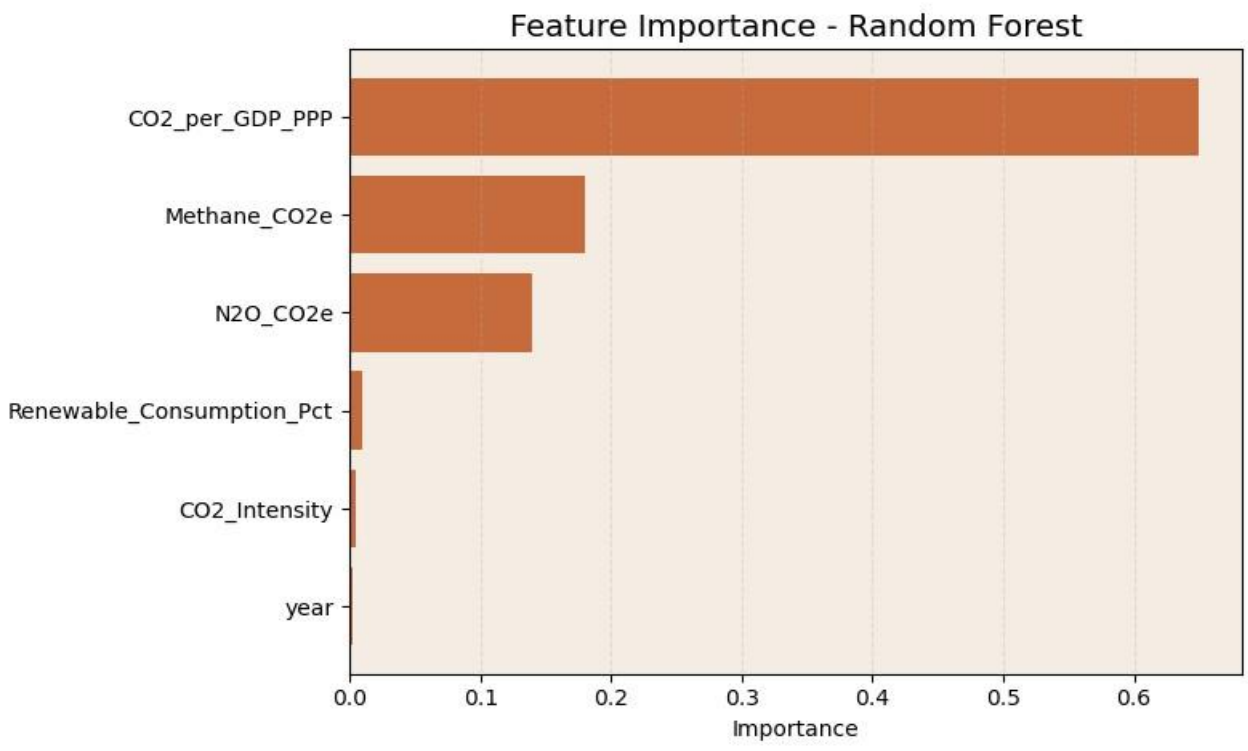


Key Insights:

I found that renewable energy is often considered one of the main solutions for reducing CO_2 emissions. When we think of lower emissions, we usually expect higher renewable energy usage to directly lead to a significant decrease.

However, according to the data, this is not entirely the case. The increase in renewable energy consumption does not strongly correlate with a decrease in CO_2 emissions. Most of the emissions are still driven by economic and industrial factors rather than energy source alone.

This suggests that while renewable energy plays a role, it is not the dominant factor in reducing emissions, and other variables such as carbon intensity and economic activity have a much stronger impact. **Feature Importance**



I found that when we think about CO_2 emissions, we often assume that renewable energy would be the most important factor in reducing emissions.

However, according to the model results, this is not the case. The feature importance graph clearly shows that **CO_2 per GDP is by far the most dominant factor**, contributing significantly more than any other variable. Greenhouse gases like **Methane (CH_4) and N_2O** also have a noticeable impact, but still much lower compared to CO_2 per GDP. On the other hand, **renewable energy consumption has very little influence**, and factors like year and CO_2 intensity contribute minimally.

This suggests that **economic efficiency and carbon intensity of economic activities play a much bigger role in driving emissions** than renewable energy alone.

Reduced Features Experiment: Role of Renewable Energy

To further evaluate the role of renewable energy in explaining CO_2 emissions, a reduced features experiment was conducted.

Goal

The objective of this experiment was to determine whether renewable energy alone can effectively explain variations in CO_2 emissions.

Methodology

In this approach, key economic indicators such as **CO_2 per GDP** and **carbon intensity** were removed from the dataset. The model was then trained using only limited features, primarily focusing on renewable energy.

Results

The results showed a significant decline in model performance:

The R^2 score dropped drastically from **0.9988 to 0.19**, this sharp decrease indicates that renewable energy alone is not sufficient to explain CO_2 emissions. It highlights the importance of economic and industrial factors in influencing emission levels.

Hypothesis Testing

To further analyze the relationship between renewable energy consumption and CO_2 emissions, hypothesis testing was conducted.

Null Hypothesis (H_0):

Renewable energy consumption has no significant relationship with CO_2 emissions per capita.

Alternative Hypothesis (H_1):

Renewable energy consumption has a significant relationship with CO_2 emissions per capita.

Findings

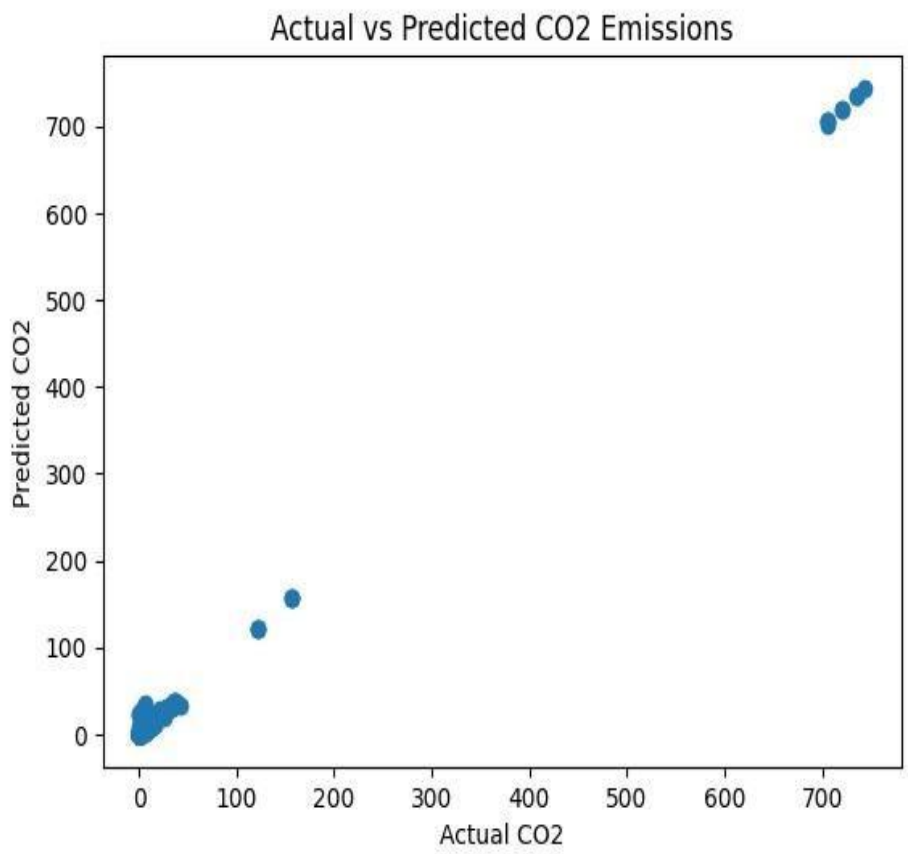
The results of the analysis showed:

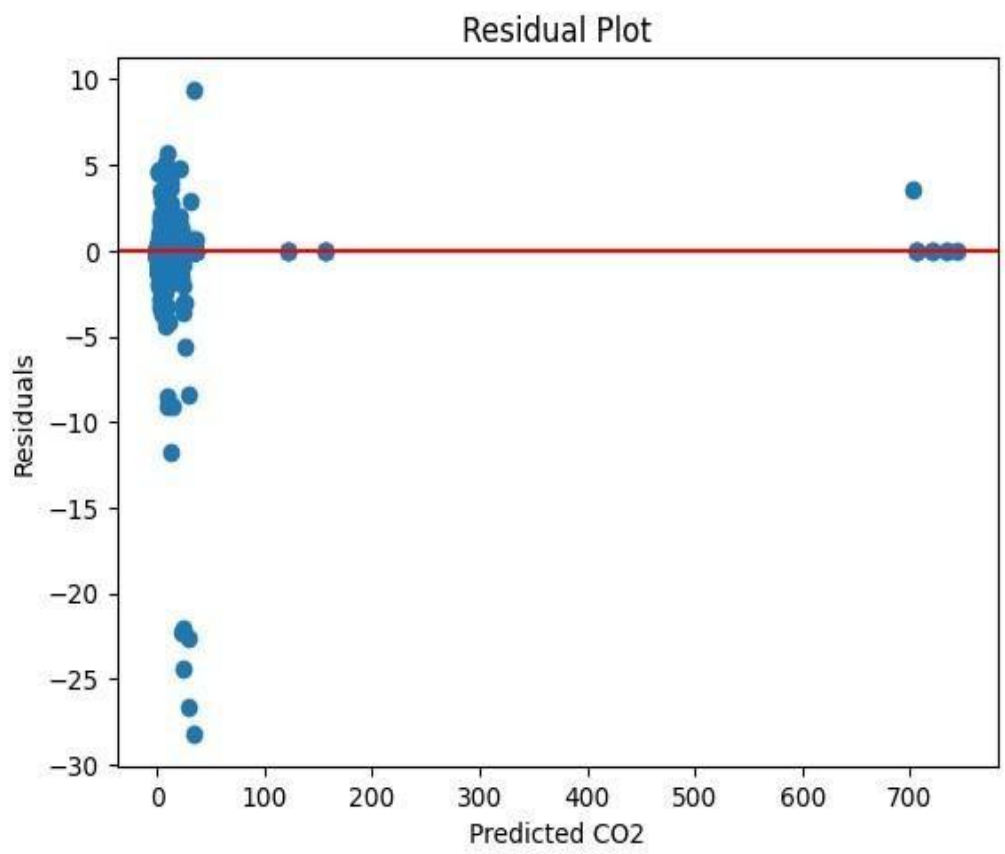
- Correlation: **-0.137** (weak negative relationship)
- p-value: $\mathbf{3.94 \times 10^{-27}}$

Interpretation

Although the p-value is extremely small, indicating that the relationship is statistically significant, the correlation value is very low. This means that while renewable energy does have some impact on CO_2 emissions, the effect is weak. Therefore, renewable energy alone cannot significantly reduce CO_2 emissions, and other factors such as economic activity and carbon intensity play a more important role.

Visual Model Evaluation





Key insights:
I found that when evaluating a model, we usually expect the predicted values to closely match the actual values if the model is performing well.
From the **Actual vs Predicted plot**, it can be observed that most of the data points lie close to the diagonal line. This indicates that the predicted CO_2 values are very close to the actual values, showing **high accuracy of the model**.
Looking at the **Residual Plot**, most of the errors are centered around zero with no clear pattern. This suggests that the model does not have significant bias and is making **balanced predictions**.
Although there are a few outliers, the overall distribution of errors is small, indicating a **good model fit**.

Hyperparameter Tuning And Optimization
To improve the performance of the Random Forest model, hyperparameter tuning was applied. After tuning, **the optimal parameters** obtained were:

- n_estimators = 200
- max_depth = None
- min_samples_split = 2
- min_samples_leaf = 1

Optimized Model Performance
The tuned model achieved the following performance:

- Cross Validation Score (CV Score): 0.9496
- R^2 Score: 0.9989
- Mean Absolute Error (MAE): 0.6377
- Root Mean Square Error (RMSE): 2.047

I found that tuning the model parameters significantly improved its performance. After optimization, the model achieved very high accuracy with reduced error values. This indicates that hyperparameter tuning helped the model generalize better on unseen data and produce more reliable predictions.

ANN Performance Improvement & Model Comparison
To further evaluate model performance, an Artificial Neural Network (ANN) model was also implemented and compared with other machine learning models.

1. ANN (Before Tuning)

- R^2 = **-0.006**
- MAE = **10.95**
- RMSE = **62.28**
- Very poor performance

2. ANN (After Tuning)• R^2 = **0.159**
• MAE = **12.24**
• RMSE = **56.93**
• Slight improvement but still low accuracy

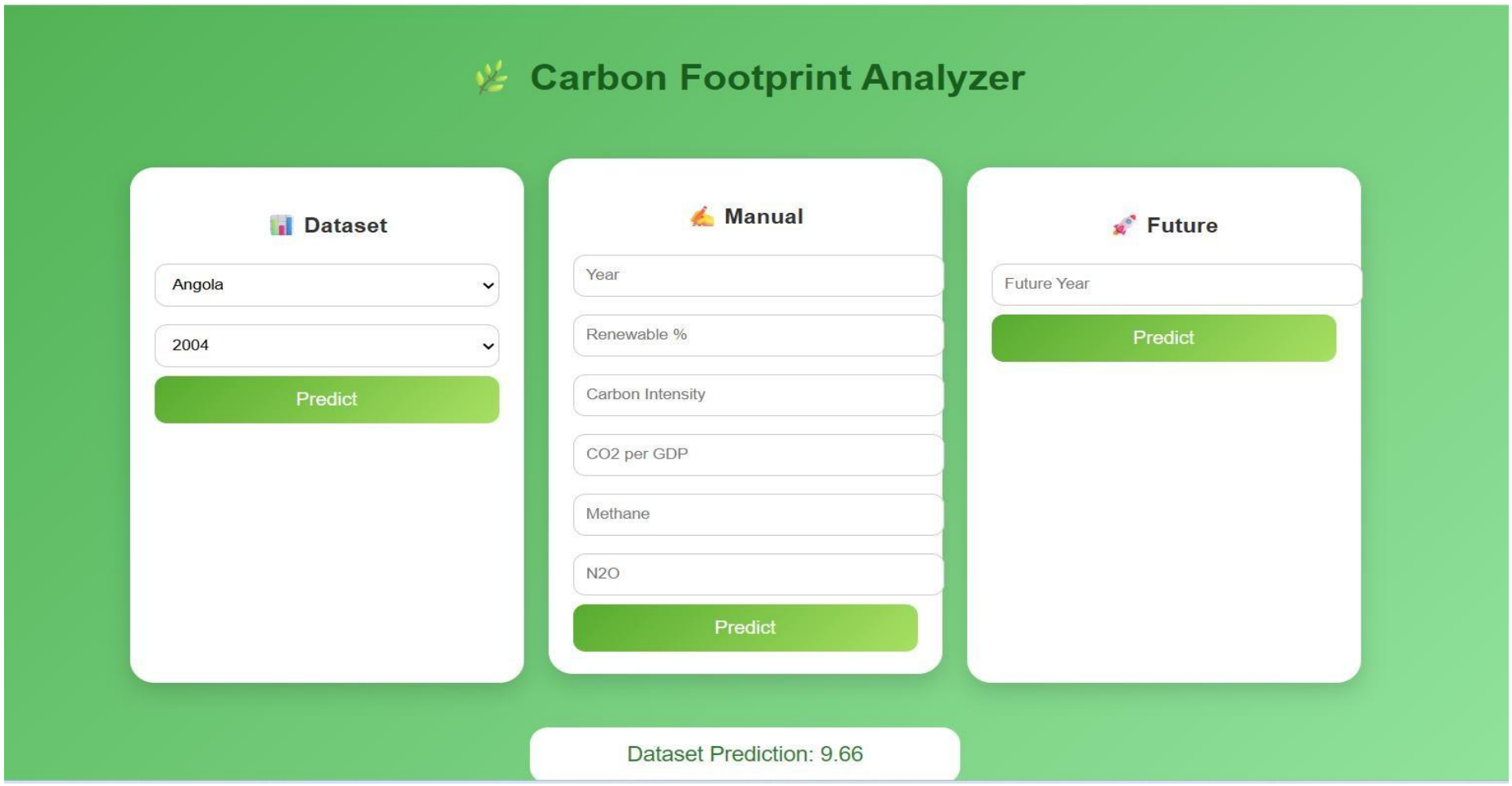
I found that when using advanced models like Artificial Neural Networks (ANN), we often expect them to perform better than traditional machine learning models.
However, in this case, the initial ANN model performed very poorly. Before tuning, the model had a negative R^2 score (**-0.006**), along with high error values, indicating that it was not able to learn the patterns in the data effectively.
After applying hyperparameter tuning, there was a slight improvement in performance. The R^2 score increased to **0.159**, and the error values decreased slightly. However, the overall accuracy remained low compared to other models.
This suggests that, despite tuning, the ANN model was not suitable for this dataset. In contrast, models like Random Forest performed significantly better, indicating that simpler ensemble methods were more effective for this problem.

Model	R^2 Score	MAE	RMSE
Linear Regression	0.568	9.22	40.79
Random Forest	0.999	0.65	2.11
ANN (Tuned)	0.159	12.24	56.93

Results and Evaluation

I found that although advanced models like ANN are often expected to perform better, the results showed that this was not the case for this dataset. The Random Forest model clearly outperformed all other models, achieving the highest accuracy with the lowest error. In contrast, Linear Regression showed only moderate performance, while ANN struggled to produce reliable predictions even after tuning.This indicates that ensemble models like Random Forest are more effective for this type of structured data, whereas more complex models like ANN may not always yield better results.

Model Deployment (Web Application)



To demonstrate the real-world application of the developed machine learning model, a web-based application was created.

The application, named **"Carbon Footprint Analyzer"**, allows users to interact with the model and generate predictions based on input values. **Features of the Application**

- Users can select country and year from the dataset
- Manual input option for: ○ Renewable energy percentage ○ Carbon intensity ○ CO_2 per GDP ○ Methane (CH_4) ○ Nitrous Oxide (N_2O)
- Future prediction option for estimating upcoming values

Functionality and Insight

The application enables users to input data and instantly receive predicted CO_2 emissions. This demonstrates that the model is not only accurate but also practical and user-friendly. Overall, this shows how machine learning models can be effectively deployed in real-world applications to support environmental analysis and decision-making.

Key Findings

Random Forest significantly outperformed Linear Regression, as it was able to capture complex non-linear patterns in the data

CO_2 per GDP is the most influential factor, highlighting the critical role of economic carbon efficiency in driving emissions

Methane (CH_4) and Nitrous Oxide (N_2O) show moderate contributions alongside economic indicators

Renewable energy has a relatively weak impact on CO_2 emissions compared to economic factors ○ **Economic carbon intensity plays a major role**, making it a key driver of emissions

Suggestions Based on Findings

Based on the analysis and findings, several recommendations can be made to address CO_2 emissions more effectively.

Firstly, since **CO_2 per GDP is the most significant factor**, efforts should be made to improve economic efficiency. Governments and industries should focus on reducing carbon intensity while maintaining economic growth.

Secondly, although renewable energy has a positive impact, its influence alone is limited. Therefore, it should be combined with other strategies such as improving industrial processes and adopting cleaner technologies.

Additionally, controlling **greenhouse gases like Methane (CH_4) and Nitrous Oxide (N_2O)** can help in reducing overall emissions, as they also contribute significantly.

Moreover, policymakers should design strategies that target **economic and industrial sectors**, as they are the major drivers of emissions rather than focusing only on renewable energy adoption.

Finally, the use of machine learning models like **Random Forest** can help in predicting emissions and assisting decision-makers in planning effective environmental policies.

Conclusion

This project analyzed the key factors influencing CO_2 emissions using data analytics and machine learning techniques. The results showed that economic factors, particularly CO_2 per GDP, have the strongest impact, while renewable energy plays a relatively smaller role.

The findings highlight that reducing carbon intensity and improving economic efficiency are essential for effectively controlling CO_2 emissions.

Future Work

Future improvements can focus on using more detailed and real-time datasets to enhance the accuracy of the analysis. Including additional variables such as policy factors, regional differences, and economic indicators could provide deeper insights into CO_2 emissions.
Advanced machine learning techniques, including deep learning models, can also be explored to improve prediction performance further.
Additionally, the web application can be enhanced by integrating real-time data and expanding its features to support more detailed environmental analysis and decision-making.

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